Sahaj Bhadja

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Summary

Systems-oriented software engineer with 2+ years using **C++** and **Python** across simulation, concurrent processing, cross-platform software, and automated validation. Experienced in debugging, failure isolation, performance optimization, testing, and maintainable object-oriented design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **distributed** Python platform with Azure SQL and Blob Storage for 500+ users, optimizing indexed queries and data access paths to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous workers processing 1,000+ jobs with explicit state transitions, error isolation, structured logging, and recovery paths, increasing concurrent throughput threefold while preventing cascading failures under production load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, integrating secure file ingestion, REST APIs, and responsive workflows while maintaining stable performance during peak load.
- Developed a **multithreaded** batch-processing pipeline for 1,000+ files using ThreadPoolExecutor, validation checks, fault-tolerant execution, progress monitoring, repeatable error handling, and status visibility, reducing end-to-end processing turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **simulation pipeline** processing 100,000+ autonomous-driving trajectory logs, improving long-horizon planning by 18% through reproducible experiments, automated validation, performance tracing, and structured model configuration.
- Implemented spatial-temporal processing that combined HD-map geometry with vehicle trajectories through multi-head attention, improving closed-loop simulation stability 15% while supporting systematic debugging and comparative analysis across complex scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform software with automated executable builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Optimized object-oriented integration modules, caching, pagination, and AWS data pipelines, improving response times 30%, increasing processing efficiency 68%, and reducing technical debt 20% through maintainable design patterns.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js capabilities across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across real-time business workflows and modular service boundaries.
- Automated testing and containerized delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages & Systems: C++, Python, Java, Bash, Shell, Linux, object-oriented programming, data structures, algorithms, operating systems, concurrency, multithreading
Development: Git, GitHub, Docker, Jenkins, Maven, CI/CD, build automation, VM-based testing, cross-platform development, technical documentation, code reviews
Testing & Debugging: PyTest, Mockito, Jest, unit testing, integration testing, regression testing, software debugging, fault isolation, error handling, logging, monitoring
Architecture & Interfaces: REST APIs, WebSockets, TCP/IP, JSON, XML, YAML, microservices, distributed systems, caching, performance optimization, simulation workflows, data validation

Projects

CPU-Scheduling Simulator | Live | *C++, React, Operating Systems, EJ-2*
- Implemented six deterministic CPU-scheduling algorithms in **C++** with React controls and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

FinNexus | Live | *Real-Time Systems, WebSockets, Kubernetes, MongoDB*
- Built a real-time platform supporting 50,000+ users and one million transactions, reliably integrating WebSocket updates, failure monitoring, Kubernetes deployment, automated releases, and performance-tuned services under high-volume workloads.